

82

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QUANT CHECKLIST

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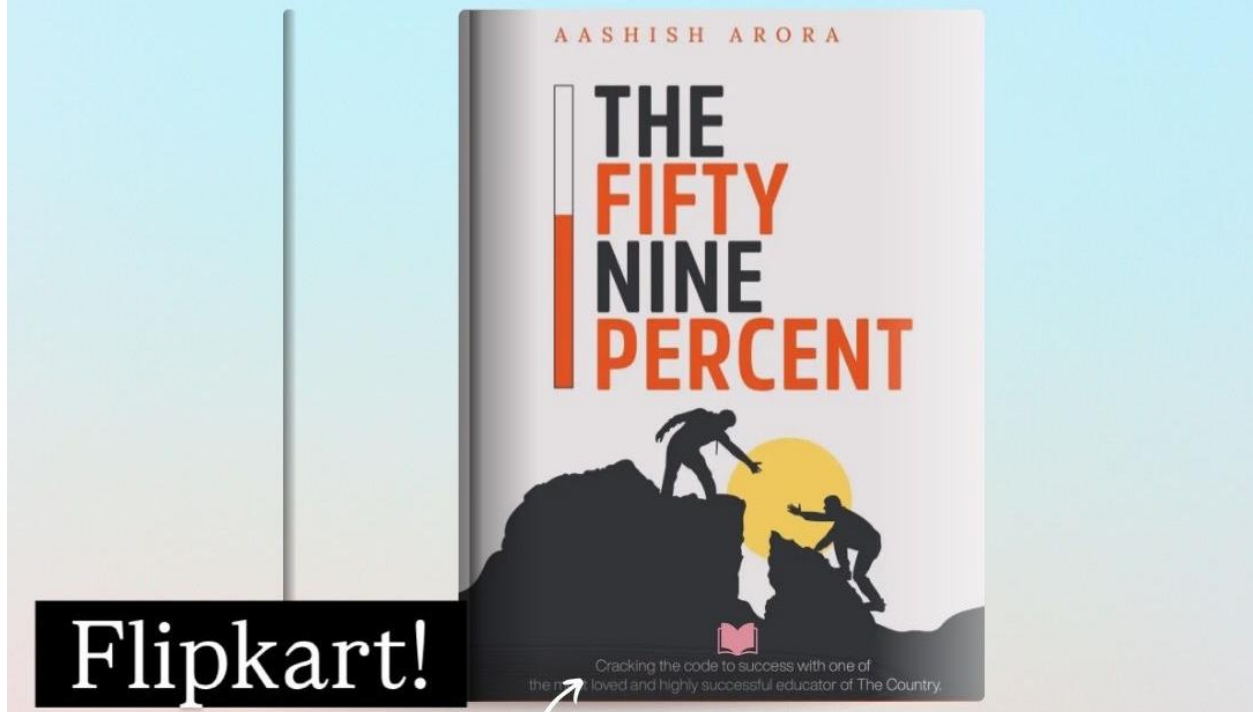


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1. SIMPLIFICATION AND APPROXIMATION

Direction: What value should come in place of the question mark (?) in the following question?

1. $? + 43.75\% \text{ of } 512 = \sqrt{6084} \times 6 - \sqrt[3]{19683}$

- A. 213
- B. 217
- C. 221
- D. 225
- E. None of these

2. $13 \times \sqrt{784} + 56.25\% \text{ of } 960 - 4^4 = ? \times 8$

- A. 77
- B. 79
- C. 81
- D. 83
- E. None of these

3. $?% \text{ of } 900 + \frac{7}{12} \text{ of } 480 = 75\% \text{ of } 840 + 2^6$

- A. 40
- B. 42
- C. 44
- D. 46
- E. None of these

4. $(49^2 - 31^2) \div 18 + 68.75\% \text{ of } 384 = ? + 15$

- A. 329
- B. 325
- C. 337
- D. 341
- E. None of these

5. $2\frac{3}{5} \text{ of } 150 - 37.5\% \text{ of } 320 + \sqrt{1444} = ? \times 4$

- A. 71
- B. 73
- C. 75
- D. 77
- E. None of these

6. $\sqrt{7056} + \sqrt[3]{59319} + 83.33\% \text{ of } 486 = ? + 390$

- A. 132
- B. 138
- C. 144
- D. 150
- E. None of these

7. $31.25\% \text{ of } 768 + \frac{7}{15} \text{ of } 450 - ? = 18^2 - 96$

- A. 222

B. 226

C. 230

D. 218

E. None of these

$$8. \left[\left(\frac{11}{16} \text{ of } 512 \right) + \left(\frac{2}{3} \text{ of } 288 \right) \right] \div 8 + 5^2 = ?$$

A. 89

B. 91

C. 93

D. 95

E. None of these

$$9. 24^2 + 27.27\% \text{ of } 363 = ?$$

A. 670

B. 675

C. 680

D. 685

E. None of these

$$10. 24 \times 36 - \frac{7}{12} \text{ of } 864 + 13^2 = ? + 275$$

A. 246

B. 250

C. 258

D. 254

E. None of these

$$11. \frac{5}{8} \text{ of } 704 - 12.5\% \text{ of } 256 + \sqrt[3]{27000} = ? \times 3$$

A. 146

B. 142

C. 138

D. 150

E. None of these

$$12. [(72 \div 9 \times 17) - (65 \div 5 \times 8)] + 41.66\% \text{ of } 360 = ?$$

A. 176

B. 180

C. 186

D. 182

E. None of these

$$13. ? \div 5 + 68.75\% \text{ of } 320 = 21^2 - 13^2$$

A. 254

B. 260

C. 266

D. 272

E. None of these

$$14. \sqrt{4096} \times 12 - \sqrt[3]{15625} \times 9 + 32.5\% \text{ of } 800 = ?$$

A. 789

B. 797

C. 803

D. 811

E. None of these

$$15. 18.75\% \text{ of } 2048 + 44.44\% \text{ of } 1080 - 6.25\% \text{ of } 960 = ? \times 6$$

A. 134

B. 130

C. 126

D. 138

E. None of these

$$16. ? - \frac{7}{9} \text{ of } 567 = 125\% \text{ of } 248 + \sqrt{2025}$$

A. 784

B. 790

C. 796

D. 804

E. None of these

$$17. [3\frac{1}{4} \text{ of } 64 + 2\frac{2}{5} \text{ of } 75] - 2^5 = ? + 290$$

A. 54

B. 58

C. 62

D. 66

E. None of these

$$18. \sqrt[3]{110592} + 14^2 + 37.5\% \text{ of } 288 = ? + 225$$

A. 123

B. 127

C. 131

D. 135

E. None of these

$$19. ?\% \text{ of } 720 - 56.25\% \text{ of } 640 = 18 \times 14$$

A. 85

B. 80

C. 90

D. 95

E. None of these

$$20. 84 \div \sqrt{196} + 1536 \div \sqrt[3]{512} + 11.11\% \text{ of } 729 = ? \times 9$$

A. 27

B. 29

C. 31

D. 33

E. None of these

Answers:

1. B

2. C

3. D

4. A

5. D

6. B

7. A

8. C

9. B

10. D

11. A

12. D

13. B

14. C

15. A

16. C

17. D

18. B

19. A

20. C

Solutions:

$$1. ? + 43.75\% \text{ of } 512 = \sqrt{6084} \times 6 - \sqrt[3]{19683}$$

$$43.75\% = \frac{7}{16}, \sqrt{6084} = 78 \text{ and } \sqrt[3]{19683} = 27$$

$$? + \frac{7}{16} \times 512 = 78 \times 6 - 27$$

$$? + 224 = 468 - 27$$

$$? + 224 = 441$$

$$? = 217$$

$$2. 13 \times \sqrt{784} + 56.25\% \text{ of } 960 - 4^4 = ? \times 8$$

$$\sqrt{784} = 28, 56.25\% = \frac{9}{16} \text{ and } 4^4 = 256$$

$$13 \times 28 + \frac{9}{16} \times 960 - 256 = ? \times 8$$

$$364 + 540 - 256 = ? \times 8$$

$$648 = ? \times 8$$

$$? = 81$$

$$3. ?\% \text{ of } 900 + \frac{7}{12} \text{ of } 480 = 75\% \text{ of } 840 + 2^6$$

$$75\% = \frac{3}{4} \text{ and } 2^6 = 64$$

$$?\% \text{ of } 900 + \frac{7}{12} \times 480 = \frac{3}{4} \times 840 + 64$$

$$?\% \text{ of } 900 + 280 = 630 + 64$$

$$?\% \text{ of } 900 = 414$$

$$? = 414 \times 100 \div 900 = 46$$

$$4. (49^2 - 31^2) \div 18 + 68.75\% \text{ of } 384 = ? + 15$$

$$68.75\% = \frac{11}{16}$$

$$(2401 - 961) \div 18 + \frac{11}{16} \times 384 = ? + 15$$

$$1440 \div 18 + 264 = ? + 15$$

$$80 + 264 = ? + 15$$

$$344 = ? + 15$$

$$? = 329$$

$$5. 2\frac{3}{5} \text{ of } 150 - 37.5\% \text{ of } 320 + \sqrt{1444} = ? \times 4$$

$$2\frac{3}{5} = \frac{13}{5}, 37.5\% = \frac{3}{8} \text{ and } \sqrt{1444} = 38$$

$$\frac{13}{5} \times 150 - \frac{3}{8} \times 320 + 38 = ? \times 4$$

$$390 - 120 + 38 = ? \times 4$$

$$308 = ? \times 4$$

$$? = 77$$

$$6. \sqrt{7056} + \sqrt[3]{59319} + 83.33\% \text{ of } 486 = ? + 390$$

$$\sqrt{7056} = 84, \sqrt[3]{59319} = 39 \text{ and } 83.33\% = \frac{5}{6}$$

$$84 + 39 + \frac{5}{6} \times 486 = ? + 390$$

$$84 + 39 + 405 = ? + 390$$

$$528 = ? + 390$$

$$? = 138$$

$$7. 31.25\% \text{ of } 768 + \frac{7}{15} \text{ of } 450 - ? = 18^2 - 96$$

$$31.25\% = \frac{5}{16} \text{ and } 18^2 = 324$$

$$\frac{5}{16} \times 768 + \frac{7}{15} \times 450 - ? = 324 - 96$$

$$240 + 210 - ? = 228$$

$$450 - ? = 228$$

$$? = 222$$

$$8. \left[\left(\frac{11}{16} \text{ of } 512 \right) + \left(\frac{2}{3} \text{ of } 288 \right) \right] \div 8 + 5^2 = ?$$

$$\frac{11}{16} \text{ of } 512 = 352, \frac{2}{3} \text{ of } 288 = 192 \text{ and } 5^2 = 25$$

$$(352 + 192) \div 8 + 25 = ?$$

$$544 \div 8 + 25 = ?$$

$$68 + 25 = ?$$

$$? = 93$$

$$9. 24^2 + 27.27\% \text{ of } 363 = ?$$

$$27.27\% = \frac{3}{11} \text{ and } 24^2 = 576$$

$$576 + \frac{3}{11} \times 363 = ?$$

$$576 + 99 = ?$$

$$? = 675$$

$$10. 24 \times 36 - \frac{7}{12} \text{ of } 864 + 13^2 = ? + 275$$

$$\frac{7}{12} \text{ of } 864 = 504 \text{ and } 13^2 = 169$$

$$864 - 504 + 169 = ? + 275$$

$$529 = ? + 275$$

$$? = 254$$

$$11. \frac{5}{8} \text{ of } 704 - 12.5\% \text{ of } 256 + \sqrt[3]{27000} = ? \times 3$$

$$12.5\% = \frac{1}{8} \text{ and } \sqrt[3]{27000} = 30$$

$$\frac{5}{8} \times 704 - \frac{1}{8} \times 256 + 30 = ? \times 3$$

$$440 - 32 + 30 = ? \times 3$$

$$438 = ? \times 3$$

$$? = 146$$

$$12. [(72 \div 9 \times 17) - (65 \div 5 \times 8)] + 41.66\% \text{ of } 360 = ?$$

$$41.66\% = \frac{5}{12}$$

$$(8 \times 17 - 13 \times 8) + \frac{5}{12} \times 360 = ?$$

$$(136 - 104) + 150 = ?$$

$$32 + 150 = ?$$

$$? = 182$$

$$13. ? \div 5 + 68.75\% \text{ of } 320 = 21^2 - 13^2$$

$$68.75\% = \frac{11}{16}$$

$$? \div 5 + \frac{11}{16} \times 320 = 441 - 169$$

$$? \div 5 + 220 = 272$$

$$? \div 5 = 52$$

$$? = 260$$

$$14. \sqrt{4096} \times 12 - \sqrt[3]{15625} \times 9 + 32.5\% \text{ of } 800 = ?$$

$$\sqrt{4096} = 64, \sqrt[3]{15625} = 25 \text{ and } 32.5\% = \frac{13}{40}$$

$$64 \times 12 - 25 \times 9 + \frac{13}{40} \times 800 = ?$$

$$768 - 225 + 260 = ?$$

$$? = 803$$

$$15. 18.75\% \text{ of } 2048 + 44.44\% \text{ of } 1080 - 6.25\% \text{ of } 960 = ? \times 6$$

$$18.75\% = \frac{3}{16}, 44.44\% = \frac{4}{9} \text{ and } 6.25\% = \frac{1}{16}$$

$$\frac{3}{16} \times 2048 + \frac{4}{9} \times 1080 - \frac{1}{16} \times 960 = ? \times 6$$

$$384 + 480 - 60 = ? \times 6$$

$$804 = ? \times 6$$

$$? = 134$$

$$16. ? - \frac{7}{9} \text{ of } 567 = 125\% \text{ of } 248 + \sqrt{2025}$$

$$125\% = \frac{5}{4} \text{ and } \sqrt{2025} = 45$$

$$? - \frac{7}{9} \times 567 = \frac{5}{4} \times 248 + 45$$

$$? - 441 = 310 + 45$$

$$? - 441 = 355$$

$$? = 796$$

$$17. [3\frac{1}{4} \text{ of } 64 + 2\frac{2}{5} \text{ of } 75] - 2^5 = ? + 290$$

$$3\frac{1}{4} = \frac{13}{4}, 2\frac{2}{5} = \frac{12}{5} \text{ and } 2^5 = 32$$

$$[\frac{13}{4} \times 64 + \frac{12}{5} \times 75] - 32 = ? + 290$$

$$[208 + 180] - 32 = ? + 290$$

$$356 = ? + 290$$

$$? = 66$$

$$18. \sqrt[3]{110592} + 14^2 + 37.5\% \text{ of } 288 = ? + 225$$

$$\sqrt[3]{110592} = 48, 14^2 = 196 \text{ and } 37.5\% = \frac{3}{8}$$

$$48 + 196 + \frac{3}{8} \times 288 = ? + 225$$

$$48 + 196 + 108 = ? + 225$$

$$352 = ? + 225$$

$$? = 127$$

$$19. ?\% \text{ of } 720 - 56.25\% \text{ of } 640 = 18 \times 14$$

$$56.25\% = \frac{9}{16} \text{ and } 18 \times 14 = 252$$

$$?\% \text{ of } 720 - \frac{9}{16} \times 640 = 252$$

$$?\% \text{ of } 720 - 360 = 252$$

$$?\% \text{ of } 720 = 612$$

$$? = 612 \times 100 \div 720 = 85$$

$$20. 84 \div \sqrt{196} + 1536 \div \sqrt[3]{512} + 11.11\% \text{ of } 729 = ? \times 9$$

$$\sqrt{196} = 14, \sqrt[3]{512} = 8 \text{ and } 11.11\% = \frac{1}{9}$$

$$84 \div 14 + 1536 \div 8 + \frac{1}{9} \times 729 = ? \times 9$$

$$6 + 192 + 81 = ? \times 9$$

$$279 = ? \times 9$$

$$? = 31$$

2. ARITHMETIC QUESTIONS

Q1: Aarohi travels from City M to City N at 60 km/hr and from City N to City O at 40 km/hr. The total distance from City M to City O is 120 km, and her average speed for the whole journey is 48 km/hr. If City M, City N and City O are in a straight line, find the distance between City M and City N.

आरोही शहर M से शहर N तक 60 किमी/घंटा की गति से और शहर N से शहर O तक 40 किमी/घंटा की गति से यात्रा करती है। शहर M से शहर O तक कुल दूरी 120 किमी है, और पूरी यात्रा के लिए उसकी औसत गति 48 किमी/घंटा है। यदि शहर M, शहर N और शहर O एक सीधी रेखा में हैं, तो शहर M और शहर N के बीच की दूरी ज्ञात कीजिए।

- (A) 50 km
- (B) 70 km
- (C) 45 km
- (D) 60 km
- (E) 80 km

Q2: 18 men can complete a work in 24 days. If 4 women can do as much work as 3 men, then find how many days 36 women will take to complete the same work.

18 पुरुष किसी कार्य को 24 दिनों में पूरा कर सकते हैं। यदि 4 महिलाएँ उतना ही कार्य कर सकती हैं जितना 3 पुरुष करते हैं, तो 36 महिलाएँ उसी कार्य को कितने दिनों में पूरा करेंगी?

- (A) 12 days
- (B) 18 days
- (C) 20 days
- (D) 16 days
- (E) 24 days

Q3: t years hence, the ratio of the ages of Kavya and Meera will be 5:6. After $(t + 8)$ years, the ratio of their ages will be 7:8. If the present age of Kavya is 14 years, find the present age of Meera.

t वर्ष बाद, काव्या और मीरा की आयु का अनुपात 5:6 होगा। $(t + 8)$ वर्ष बाद, उनकी आयु का अनुपात 7:8 होगा। यदि काव्या की वर्तमान आयु 14 वर्ष है, तो मीरा की वर्तमान आयु ज्ञात कीजिए।

- (A) 20 years
- (B) 18 years
- (C) 16 years

(D) 24 years

(E) 21 years

Q4: A 360-meter-long train crosses a 240-meter-long bridge in 20 seconds. Find the time taken by the train to cross a man running in the opposite direction at 15 m/sec.

360 मीटर लंबी एक ट्रेन 240 मीटर लंबे पुल को 20 सेकंड में पार करती है। 15 मीटर/सेकंड की गति से विपरीत दिशा में दौड़ रहे एक व्यक्ति को पार करने में ट्रेन द्वारा लिया गया समय ज्ञात कीजिए।

(A) 6 seconds

(B) 10 seconds

(C) 12 seconds

(D) 15 seconds

(E) 8 seconds

Q5: The average age of 3 sons in a family is 10 years, and the average age of the 3 sons together with their parents is 18 years. If the mother is 6 years younger than the father, find the age of the father 3 years ago.

एक परिवार में 3 बेटों की औसत आयु 10 वर्ष है, और 3 बेटों तथा उनके माता-पिता की कुल औसत आयु 18 वर्ष है। यदि माता, पिता से 6 वर्ष छोटी है, तो 3 वर्ष पहले पिता की आयु ज्ञात कीजिए।

(A) 30 years

(B) 28 years

(C) 32 years

(D) 34 years

(E) 36 years

Q6: If a person increases his speed from 50 km/hr to 75 km/hr, he takes 40 minutes less to cover a certain distance D. Find the time taken by him to cover 180% of D at his original speed. Give the answer in minutes.

यदि कोई व्यक्ति अपनी चाल 50 किमी/घंटा से बढ़ाकर 75 किमी/घंटा कर देता है, तो वह एक निश्चित दूरी D तय करने में 40 मिनट कम समय लेता है। उसकी मूल चाल से D की 180% दूरी तय करने में लगने वाला समय मिनटों में ज्ञात कीजिए।

(A) 180 minutes

(B) 200 minutes

(C) 216 minutes

(D) 240 minutes

(E) 225 minutes

Q7: The sum of two numbers is equal to the sum of the square of 13 and the cube of 7. The larger number is 40 less than the square of 22. Find the sum of three times 25% of the smaller number and half of the larger number.

दो संख्याओं का योग 13 के वर्ग और 7 के घन के योग के बराबर है। बड़ी संख्या 22 के वर्ग से 40 कम है। छोटी संख्या के 25% के तीन गुने और बड़ी संख्या के आधे का योग ज्ञात कीजिए।

- (A) 253
- (B) 263
- (C) 273
- (D) 283
- (E) 293

Q8: The length of a rectangular park is 160% of the radius of a circular park whose circumference is 220 meters. If the area of the rectangular park is 1568 square meters, then the radius of the circular park is what percent more than the breadth of the rectangular park? Take $\pi = 22/7$.

एक आयताकार पार्क की लंबाई एक वृत्ताकार पार्क की त्रिज्या की 160% है, जिसकी परिधि 220 मीटर है। यदि आयताकार पार्क का क्षेत्रफल 1568 वर्ग मीटर है, तो वृत्ताकार पार्क की त्रिज्या आयताकार पार्क की चौड़ाई से कितने प्रतिशत अधिक है? $\pi = 22/7$ लीजिए।

- (A) 20%
- (B) 25%
- (C) 30%
- (D) 40%
- (E) 50%

Q9: A jar contains 90 ml of a mixture of two liquids A and B. Liquid A is 50 ml more than liquid B. If x ml of liquid A is taken out from the jar, the ratio of liquid A to liquid B in the remaining mixture becomes 5 : 2. Find x.

एक जार में दो द्रव A और B का 90 मिली मिश्रण है। द्रव A, द्रव B से 50 मिली अधिक है। यदि जार से द्रव A के x मिली निकाल दिए जाएँ, तो शेष मिश्रण में द्रव A और द्रव B का अनुपात 5 : 2 हो जाता है। x ज्ञात कीजिए।

- (A) 12 ml
- (B) 15 ml
- (C) 18 ml
- (D) 20 ml

(E) 25 ml

Q10: The ratio of the ages of Ritu and Neha after 6 years will be 5 : 8. The ratio of the present ages of Komal and Neha is 7 : 9. If the sum of the ages of Ritu and Komal after 4 years will be 132 years, find the sum of the present ages of Ritu and Neha.

6 वर्ष बाद रितु और नेहा की आयु का अनुपात 5 : 8 होगा। कोमल और नेहा की वर्तमान आयु का अनुपात 7 : 9 है। यदि 4 वर्ष बाद रितु और कोमल की आयु का योग 132 वर्ष होगा, तो रितु और नेहा की वर्तमान आयु का योग ज्ञात कीजिए।

- (A) 120 years
- (B) 132 years
- (C) 144 years
- (D) 156 years
- (E) 168 years

Q11: Prateek can cover 1 km in 8 minutes, and Yogesh can cover 1 km in 12 minutes. If both participate in a race and Prateek defeats Yogesh by 300 meters, find the length of the race.

प्रतीक 1 किमी की दूरी 8 मिनट में तय कर सकता है और योगेश 1 किमी की दूरी 12 मिनट में तय कर सकता है। यदि दोनों एक दौड़ में भाग लेते हैं और प्रतीक, योगेश को 300 मीटर से हरा देता है, तो दौड़ की लंबाई ज्ञात कीजिए।

- (A) 700 m
- (B) 800 m
- (C) 900 m
- (D) 1000 m
- (E) 600 m

Q12: Keshav can complete a piece of work in 240 days. Varun is 50% more efficient than Keshav. Keshav and Tejas together can complete the same work in 160 days. If all three work together, in how many days will the work be completed?

केशव किसी काम को 240 दिनों में पूरा कर सकता है। वरुण, केशव से 50% अधिक दक्ष है। केशव और तेजस मिलकर उसी काम को 160 दिनों में पूरा कर सकते हैं। यदि तीनों मिलकर काम करें, तो काम कितने दिनों में पूरा होगा?

- (A) 70 days
- (B) 90 days
- (C) 96 days

- (D) 80 days
- (E) 120 days

Q13: The cost price of 12 pens is equal to the marked price of 9 pens. When a trader sells one pen at a 25% discount, he earns Rs. 80 less than the amount he earns when he sells it at a 15% discount. Find the cost price of one pen.

12 पेन का क्रय मूल्य 9 पेन के अंकित मूल्य के बराबर है। जब एक व्यापारी एक पेन को 25% छूट पर बेचता है, तो वह 15% छूट पर बेचने की तुलना में Rs. 80 कम कमाता है। एक पेन का क्रय मूल्य ज्ञात कीजिए।

- (A) Rs. 600
- (B) Rs. 500
- (C) Rs. 640
- (D) Rs. 720
- (E) Rs. 560

Q14: A spherical metal ball is melted to make 128 identical cylindrical rods. The radius of each cylindrical rod is 25% of the radius of the spherical ball, and the height of each rod is 2 cm less than its radius. Find the radius of the spherical ball.

एक गोलाकार धातु की गेंद को पिघलाकर 128 समान बेलनाकार छड़ें बनाई जाती हैं। प्रत्येक बेलनाकार छड़ की त्रिज्या, गोलाकार गेंद की त्रिज्या का 25% है और प्रत्येक छड़ की ऊंचाई उसकी त्रिज्या से 2 सेमी कम है। गोलाकार गेंद की त्रिज्या ज्ञात कीजिए।

- (A) 18 cm
- (B) 20 cm
- (C) 24 cm
- (D) 28 cm
- (E) 32 cm

Q15: Vivek purchased 6 kg rice and 3 kg wheat for Rs. 330. Naren purchased 4 kg rice and 5 kg wheat for Rs. 310. Find the per kg price of wheat as a percentage of the per kg price of rice.

विवेक ने 6 किग्रा चावल और 3 किग्रा गेहूं Rs. 330 में खरीदा। नरेन ने 4 किग्रा चावल और 5 किग्रा गेहूं Rs. 310 में खरीदा। गेहूं का प्रति किग्रा मूल्य, चावल के प्रति किग्रा मूल्य का कितना प्रतिशत है?

- (A) 60%
- (B) 75%
- (C) 80%
- (D) 90%

(E) 65%

Q16: The area of a square is 784 cm^2 , and the perimeter of the square is equal to the perimeter of a rectangle. If the length of the rectangle is 25% more than the side of the square, then find the area of the rectangle.

एक वर्ग का क्षेत्रफल 784 सेमी^2 है और वर्ग का परिमाण एक आयत के परिमाण के बराबर है। यदि आयत की लंबाई वर्ग की भुजा से 25% अधिक है, तो आयत का क्षेत्रफल ज्ञात कीजिए।

(A) 735 cm^2 (B) 720 cm^2 (C) 756 cm^2 (D) 784 cm^2 (E) 805 cm^2

Q17: Harsh is 50% more efficient than Manish, who takes 25% more time than Suresh to complete the same work. If Harsh and Suresh together can complete 55% of the work in 10 days, then in how many days can Manish and Suresh together complete 90% of the work?

हर्ष, मनीष से 50% अधिक कुशल है, और मनीष उसी कार्य को पूरा करने में सुरेश से 25% अधिक समय लेता है। यदि हर्ष और सुरेश मिलकर 55% कार्य 10 दिनों में पूरा कर सकते हैं, तो मनीष और सुरेश मिलकर 90% कार्य कितने दिनों में पूरा करेंगे?

(A) 16 days

(B) 18 days

(C) 20 days

(D) 22 days

(E) 25 days

Q18: The length of train M is 240 meters, and train N is 'a' meters longer than train M. Both trains run at the same speed and cross a signal pole in 12 seconds and 18 seconds, respectively. Find the time taken by train N to cross a 300-meter-long platform.

ट्रेन M की लंबाई 240 मीटर है और ट्रेन N, ट्रेन M से 'a' मीटर लंबी है। दोनों ट्रेनें समान गति से चलती हैं और क्रमशः 12 सेकंड तथा 18 सेकंड में एक सिग्नल पोल को पार करती हैं। ट्रेन N द्वारा 300 मीटर लंबे प्लेटफॉर्म को पार करने में लिया गया समय ज्ञात कीजिए।

(A) 30 sec

(B) 36 sec

(C) 28 sec

(D) 33 sec

(E) 40 sec

Q19: A person invested Rs. $4P$ at 10% compound interest per annum, compounded annually, for 2 years and Rs. P at 30% simple interest per annum for 2 years. If the difference between the compound interest and the simple interest received after 2 years is Rs. 3600, then find the value of P .

एक व्यक्ति ने Rs. $4P$ को 2 वर्षों के लिए 10% वार्षिक चक्रवृद्धि ब्याज दर पर, वार्षिक संयोजन के साथ, और Rs. P को 2 वर्षों के लिए 30% वार्षिक साधारण ब्याज दर पर निवेश किया। यदि 2 वर्षों के बाद प्राप्त चक्रवृद्धि ब्याज और साधारण ब्याज के बीच का अंतर Rs. 3600 है, तो P का मान ज्ञात कीजिए।

- (A) Rs. 12000
- (B) Rs. 18000
- (C) Rs. 20000
- (D) Rs. 10000
- (E) Rs. 15000

Q20: If the upstream speed of a boat is reduced by 20%, it becomes equal to the speed of the current. If the speed of the boat in still water is 90 km/hr, then find the downstream speed of the boat.

यदि एक नाव की धारा के प्रतिकूल गति में 20% की कमी की जाए, तो वह धारा की गति के बराबर हो जाती है। यदि शांत जल में नाव की गति 90 km/hr है, तो नाव की धारा के अनुकूल गति ज्ञात कीजिए।

- (A) 110 km/hr
- (B) 120 km/hr
- (C) 130 km/hr
- (D) 140 km/hr
- (E) 150 km/hr

Answers

- | | |
|------|-------|
| 1. D | 6. C |
| 2. D | 7. C |
| 3. B | 8. B |
| 4. E | 9. D |
| 5. A | 10. C |

11.C
12.D
13.A
14.C
15.B

16.A
17.C
18.D
19.E
20.C

Solutions:

1. Option D

Let the distance between City M and City N be x km.

Then the distance between City N and City O = $120 - x$ km.

Average speed for the whole journey = 48 km/hr.

Total distance = 120 km.

Total time taken = $120/48 = 5/2$ hours.

Time taken from M to N = $x/60$ hours.

Time taken from N to O = $(120 - x)/40$ hours.

So, $x/60 + (120 - x)/40 = 5/2$

$x/60 + 120/40 - x/40 = 5/2$

$x/60 + 3 - x/40 = 5/2$

$3 - 5/2 = x/40 - x/60$

$1/2 = x \times (3 - 2)/120$

$1/2 = x/120$

$x = 60$

2. Option D

18 men can complete the work in 24 days.

Total work = $18 \times 24 = 432$ man-days

Given, 4 women can do the work of 3 men.

So, 36 women can do the work of 27 men.

Now, 27 men equivalent workers will complete 432 man-days of work.

Required time = $432/27$

Required time = 16 days

3. Option B

t years hence, the age ratio of Kavya and Meera will be 5:6.

Let Kavya's age after t years = $5x$.

Let Meera's age after t years = $6x$.

After 8 more years, their ages will be $5x + 8$ and $6x + 8$.

Given, after (t + 8) years, the ratio becomes 7:8.

So, $(5x + 8)/(6x + 8) = 7/8$

Cross multiply:

$$8 \times (5x + 8) = 7 \times (6x + 8)$$

$$40x + 64 = 42x + 56$$

$$64 - 56 = 42x - 40x$$

$$8 = 2x$$

$$x = 4$$

Kavya's age after t years = $5x = 5 \times 4 = 20$ years.

Kavya's present age = 14 years.

So, $t = 20 - 14 = 6$ years.

Meera's age after t years = $6x = 6 \times 4 = 24$ years.

Meera's present age = $24 - 6 = 18$ years.

4. Option E

Length of train = 360 meters.

Length of bridge = 240 meters.

When a train crosses a bridge, distance covered = train length + bridge length.

Total distance covered = $360 + 240 = 600$ meters.

Time taken = 20 seconds.

Speed of train = $600/20$

Speed of train = 30 m/sec.

The man is running in the opposite direction at 15 m/sec.

Relative speed = $30 + 15 = 45$ m/sec.

When a train crosses a man, distance covered = length of train.

Time taken = $360/45$

Time taken = 8 seconds.

5. Option A

Average age of 3 sons = 10 years.

Total age of 3 sons = $3 \times 10 = 30$ years.

Average age of 3 sons and their parents = 18 years.

Total number of persons = $3 + 2 = 5$.

Total age of all 5 persons = $5 \times 18 = 90$ years.

Total age of father and mother = $90 - 30 = 60$ years.

Let father's present age be F years.

Mother is 6 years younger than father.

So, mother's present age = $F - 6$.

Now, $F + (F - 6) = 60$

$2F - 6 = 60$

$2F = 66$

$F = 33$ years

Father's age 3 years ago = $33 - 3$

Father's age 3 years ago = 30 years.

6. Option C

Original speed = 50 km/hr

Increased speed = 75 km/hr

Time saved = 40 minutes = $40/60 = 2/3$ hour

Let the distance be D km.

Time taken at original speed = $D/50$

Time taken at increased speed = $D/75$

According to the question:

$D/50 - D/75 = 2/3$

LCM of 50 and 75 = 150

$3D/150 - 2D/150 = 2/3$

$D/150 = 2/3$

$D = 150 \times 2/3$

$D = 100$ km

Now, 180% of $D = 100 \times 180/100 = 180$ km

Time taken at original speed = $180/50 = 18/5$ hours

$18/5$ hours = $18/5 \times 60 = 216$ minutes

7. Option C

Sum of two numbers = Square of 13 + Cube of 7

Square of 13 = $13 \times 13 = 169$

Cube of 7 = $7 \times 7 \times 7 = 343$

Sum of two numbers = $169 + 343 = 512$

Larger number = Square of 22 – 40

Larger number = $22 \times 22 - 40$

$$\text{Larger number} = 484 - 40 = 444$$

$$\text{Smaller number} = 512 - 444 = 68$$

$$25\% \text{ of smaller number} = 68 \times 25/100 = 17$$

$$\text{Three times } 25\% \text{ of smaller number} = 3 \times 17 = 51$$

$$\text{Half of larger number} = 444/2 = 222$$

$$\text{Required sum} = 51 + 222 = 273$$

8. Option B

$$\text{Circumference of circular park} = 220 \text{ meters}$$

$$\text{Formula of circumference} = 2 \times \pi \times \text{radius}$$

$$\text{Using } \pi = 22/7:$$

$$\text{Radius} = 220 \times 7/(2 \times 22)$$

$$\text{Radius} = 220 \times 7/44$$

$$\text{Radius} = 5 \times 7 = 35 \text{ meters}$$

$$\text{Length of rectangular park} = 160\% \text{ of radius}$$

$$\text{Length} = 35 \times 160/100 = 56 \text{ meters}$$

$$\text{Area of rectangular park} = 1568 \text{ square meters}$$

$$\text{Area of rectangle} = \text{Length} \times \text{Breadth}$$

$$\text{Breadth} = \text{Area}/\text{Length}$$

$$\text{Breadth} = 1568/56 = 28 \text{ meters}$$

$$\text{Radius} = 35 \text{ meters}$$

$$\text{Breadth} = 28 \text{ meters}$$

$$\text{Difference} = 35 - 28 = 7 \text{ meters}$$

$$\text{Required percentage more} = \text{Difference} \times 100/\text{Breadth}$$

$$\text{Required percentage more} = 7 \times 100/28 = 25\%$$

9. Option D

$$\text{Total mixture} = 90 \text{ ml}$$

$$\text{Let the quantity of liquid B be } y \text{ ml.}$$

$$\text{Liquid A is } 50 \text{ ml more than liquid B.}$$

$$\text{So, quantity of liquid A} = y + 50$$

Now,

$$y + y + 50 = 90$$

$$2y + 50 = 90$$

$$2y = 90 - 50$$

$$2y = 40$$

$$y = 40/2 = 20$$

Quantity of liquid B = 20 ml

Quantity of liquid A = 20 + 50 = 70 ml

Now, x ml of liquid A is taken out.

Remaining quantity of liquid A = 70 - x

Quantity of liquid B remains = 20 ml

New ratio of liquid A to liquid B = 5 : 2

So, $(70 - x)/20 = 5/2$

Cross multiply:

$$2(70 - x) = 5 \times 20$$

$$140 - 2x = 100$$

$$2x = 140 - 100$$

$$2x = 40$$

$$x = 40/2 = 20$$

10. Option C

After 6 years, the ratio of ages of Ritu and Neha will be 5 : 8.

Let their ages after 6 years be 5x and 8x.

Present age of Ritu = 5x - 6

Present age of Neha = 8x - 6

Present age ratio of Komal and Neha = 7 : 9

So, present age of Komal = 7/9 of Neha's present age

Present age of Komal = $7/9 \times (8x - 6)$

After 4 years, sum of ages of Ritu and Komal = 132 years.

So, present sum of their ages = $132 - 8 = 124$ years

Now,

$$(5x - 6) + 7/9 \times (8x - 6) = 124$$

Multiply by 9:

$$9(5x - 6) + 7(8x - 6) = 124 \times 9$$

$$45x - 54 + 56x - 42 = 1116$$

$$101x - 96 = 1116$$

$$101x = 1116 + 96$$

$$101x = 1212$$

$$x = 1212/101 = 12$$

Present age of Ritu = $5 \times 12 - 6 = 54$ years

Present age of Neha = $8 \times 12 - 6 = 90$ years

Required sum = $54 + 90 = 144$ years

11. Option C

Prateek covers 1 km in 8 minutes.

Yogesh covers 1 km in 12 minutes.

For the same distance, speed is inversely proportional to time.

So, speed ratio of Prateek and Yogesh = $12 : 8 = 3 : 2$.

When Prateek finishes the race, Yogesh covers $\frac{2}{3}$ of the race distance.

So, Prateek's lead = $1 - \frac{2}{3} = \frac{1}{3}$ of the race distance.

Given lead = 300 meters.

So, $\frac{1}{3}$ of race distance = 300.

Race distance = $300 \times 3 = 900$ meters.

12. Option D

Take total work as 480 units.

Keshav completes the work in 240 days.

Keshav's efficiency = $480/240 = 2$ units per day.

Varun is 50% more efficient than Keshav.

Varun's efficiency = $150/100 \times 2 = 3$ units per day.

Keshav and Tejas together complete the work in 160 days.

Their combined efficiency = $480/160 = 3$ units per day.

Tejas's efficiency = $3 - 2 = 1$ unit per day.

Total efficiency of Keshav, Varun and Tejas = $2 + 3 + 1 = 6$ units per day.

Time taken by all three together = $480/6 = 80$ days.

13. Option A

Let the cost price of one pen be Rs. C.

Let the marked price of one pen be Rs. M.

Cost price of 12 pens = Marked price of 9 pens.

So,

$$12C = 9M$$

$$C = 9M/12$$

$$C = 3M/4.$$

Difference between 25% discount and 15% discount = 10% of marked price.

This difference in selling price is Rs. 80.

So,

$$10\% \text{ of } M = 80$$

$$10M/100 = 80$$

$$M = 80 \times 100/10$$

$$M = 800.$$

Cost price of one pen = $3M/4$.

Cost price = $3 \times 800/4 = \text{Rs. } 600$.

14. Option C

Let the radius of the spherical ball be R cm.

Radius of each cylindrical rod = 25% of R .

Radius of each cylindrical rod = $R/4$.

Height of each cylindrical rod = $R/4 - 2$.

Volume of spherical ball = $\frac{4}{3} \times \pi \times R \times R \times R$.

Volume of 128 cylindrical rods = $128 \times \pi \times R/4 \times R/4 \times (R/4 - 2)$.

Since the sphere is melted to form the rods:

$$\frac{4}{3} \times \pi \times R \times R \times R = 128 \times \pi \times R/4 \times R/4 \times (R/4 - 2)$$

Cancel π and $R \times R$ from both sides:

$$4R/3 = 128/16 \times (R/4 - 2)$$

$$4R/3 = 8(R/4 - 2)$$

$$4R/3 = 2R - 16$$

Multiply by 3:

$$4R = 6R - 48$$

$$6R - 4R = 48$$

$$2R = 48$$

$$R = 48/2$$

$$R = 24.$$

15. Option B

Let the price of rice be Rs. R per kg.

Let the price of wheat be Rs. W per kg.

According to the first condition:

$$6R + 3W = 330$$

Divide by 3:

$$2R + W = 110$$

$$\text{So, } W = 110 - 2R.$$

According to the second condition:

$$4R + 5W = 310$$

Put $W = 110 - 2R$:

$$4R + 5(110 - 2R) = 310$$

$$4R + 550 - 10R = 310$$

$$550 - 6R = 310$$

$$6R = 550 - 310$$

$$6R = 240$$

$$R = 240/6$$

$$R = 40.$$

$$\text{Now, } W = 110 - 2R.$$

$$W = 110 - 2 \times 40 = 30.$$

Price of wheat = Rs. 30 per kg.

Price of rice = Rs. 40 per kg.

$$\text{Required percentage} = 30/40 \times 100.$$

$$\text{Required percentage} = 75\%.$$

16. Option A

$$\text{Area of square} = 784 \text{ cm}^2$$

$$\text{Side of square} = 28 \text{ cm, because } 28 \times 28 = 784$$

$$\text{Perimeter of square} = 4 \times 28 = 112 \text{ cm}$$

$$\text{Perimeter of rectangle} = 112 \text{ cm}$$

$$\text{So, } 2 \times (\text{Length} + \text{Breadth}) = 112$$

$$\text{Length} + \text{Breadth} = 112/2 = 56 \text{ cm}$$

Length of rectangle is 25% more than the side of square.

$$\text{Length} = 28 + 25 \times 28/100$$

$$\text{Length} = 28 + 7$$

$$\text{Length} = 35 \text{ cm}$$

$$\text{Breadth} = 56 - 35$$

$$\text{Breadth} = 21 \text{ cm}$$

$$\text{Area of rectangle} = \text{Length} \times \text{Breadth}$$

$$\text{Area of rectangle} = 35 \times 21$$

$$\text{Area of rectangle} = 735 \text{ cm}^2$$

17. Option C

Let Manish's efficiency be 8 units per day.

Harsh is 50% more efficient than Manish.

$$\text{Harsh's efficiency} = 8 + 50 \times 8/100$$

$$\text{Harsh's efficiency} = 8 + 4 = 12 \text{ units per day}$$

Manish takes 25% more time than Suresh.

$$\text{So, Manish's time : Suresh's time} = 125 : 100 = 5 : 4$$

Efficiency is inverse of time.

$$\text{So, Manish's efficiency : Suresh's efficiency} = 4 : 5$$

Manish's efficiency = 8 units per day

Suresh's efficiency = 10 units per day

Harsh and Suresh's combined efficiency = $12 + 10 = 22$ units per day

Work done by Harsh and Suresh in 10 days = $22 \times 10 = 220$ units

Given, 55% of total work = 220 units

Total work = $220 \times 100/55$

Total work = 400 units

Manish and Suresh's combined efficiency = $8 + 10 = 18$ units per day

90% of total work = $90 \times 400/100$

90% of total work = 360 units

Required time = $360/18$

Required time = 20 days

18. Option D

Length of train M = 240 meters

Train M crosses a signal pole in 12 seconds.

Speed of train M = $240/12$

Speed of train M = 20 m/s

Both trains run at the same speed.

So, speed of train N = 20 m/s

Train N crosses a signal pole in 18 seconds.

Length of train N = 20×18

Length of train N = 360 meters

Length of platform = 300 meters

Total distance covered by train N to cross the platform = $360 + 300$

Total distance = 660 meters

Time taken = $660/20$

Time taken = 33 seconds

19. Option E

Amount invested at compound interest = Rs. 4P

Rate of compound interest = 10% per annum

Time = 2 years

Compound interest for 2 years at 10% = 21% of principal

Compound interest = $21 \times 4P/100$

Compound interest = $84P/100$

Amount invested at simple interest = Rs. P

Rate of simple interest = 30% per annum

Time = 2 years

Simple interest = $P \times 30 \times 2/100$

Simple interest = $60P/100$

Difference between compound interest and simple interest = Rs. 3600

So:

$84P/100 - 60P/100 = 3600$

$24P/100 = 3600$

$24P = 3600 \times 100$

$P = 360000/24$

$P = 15000$

20. Option C

Let the upstream speed of the boat be x km/hr.

If upstream speed is reduced by 20%, it becomes equal to the speed of the current.

Speed of current = 80% of x

Speed of current = $80x/100 = 4x/5$

Speed in still water = Upstream speed + Speed of current

Given speed in still water = 90 km/hr

So:

$x + 4x/5 = 90$

$9x/5 = 90$

$9x = 450$

$x = 50$

Upstream speed = 50 km/hr

Speed of current = $4 \times 50/5$

Speed of current = 40 km/hr

Downstream speed = Speed in still water + Speed of current

Downstream speed = $90 + 40$

Downstream speed = 130 km/hr

3. Quadratic Equations

CHECKLIST

In each of the following questions, there are two equations. You have to solve both equations and mark the correct answer.

(a) $x > y$

(b) $x < y$

(c) $x = y$ or the relationship cannot be established

(d) $x \geq y$

(e) $x \leq y$

1.) I. $4x^2 - 46x + 126 = 0$

II. $5y^2 - 18y + 16 = 0$

2.) I. $6x^2 - 26x + 28 = 0$

II. $7y^2 - 41y + 60 = 0$

3.) I. $8x^2 - 70x + 138 = 0$

II. $9y^2 - 44y + 51 = 0$

4.) I. $x^2 - 9\sqrt{3}x + 42 = 0$

II. $y^2 - 2\sqrt{3}y + 3 = 0$

5.) I. $10x^2 - 46x + 48 = 0$

II. $11y^2 - 51y + 54 = 0$

6.) I. $12x^2 - 44x + 40 = 0$

II. $13y^2 - 65y + 78 = 0$

7.) I. $14x^2 - 72x + 90 = 0$

II. $15y^2 - 55y + 50 = 0$

8.) I. $16x^2 - 76x + 88 = 0$

II. $5y^2 - 39y + 70 = 0$

9.) I. $7x^2 - 44x + 69 = 0$

II. $10y^2 - 45y + 45 = 0$

10.) I. $9x^2 - 51x + 60 = 0$

II. $12y^2 - 45y + 42 = 0$

11.) I. $x^2 - 3\sqrt{7}x + 14 = 0$

II. $y^2 - 9\sqrt{7}y + 98 = 0$

12.) I. $11x^2 - 56x + 69 = 0$

II. $4y^2 - 26y + 42 = 0$

13.) I. $5x^2 - 45x + 100 = 0$

II. $8y^2 - 32y + 30 = 0$

14.) I. $6x^2 - 27x + 27 = 0$

II. $9y^2 - 51y + 66 = 0$

15.) I. $13x^2 - 78x + 104 = 0$

II. $14y^2 - 50y + 44 = 0$

16.) I. $15x^2 - 54x + 48 = 0$

II. $16y^2 - 88y + 120 = 0$

17.) I. $x^2 - 7\sqrt{2}x + 12 = 0$

II. $y^2 - 9\sqrt{2}y + 28 = 0$

18.) I. $10x^2 - 38x + 36 = 0$

II. $7y^2 - 47y + 66 = 0$

19.) I. $12x^2 - 90x + 162 = 0$

II. $11y^2 - 51y + 58 = 0$

20.) I. $8x^2 - 36x + 40 = 0$

II. $6y^2 - 34y + 40 = 0$

Answers:

1. A

2. B

3. D

4. A

5. C

6. E

7. A

8. B

9. D

10. C

11. E

12. E

13. A

14. C

15. D

16. B

17. C

18. E

19. A

20. C

Solutions:

$$(1) x = 7, \frac{9}{2}$$

$$y = 2, \frac{8}{5}$$

$$(2) x = \frac{7}{3}, 2$$

$$y = 3, \frac{20}{7}$$

$$(3) x = \frac{23}{4}, 3$$

$$y = 3, \frac{17}{9}$$

$$(4) x = 7\sqrt{3}, 2\sqrt{3}$$

$$y = \sqrt{3}, \sqrt{3}$$

$$(5) x = 3, \frac{8}{5}$$

$$y = 3, \frac{18}{11}$$

$$(6) x = 2, \frac{5}{3}$$

$$y = 3, 2$$

$$(7) x = 3, \frac{15}{7}$$

$$y = 2, \frac{5}{3}$$

$$(8) x = \frac{11}{4}, 2$$

$$y = 5, \frac{14}{5}$$

$$(9) x = \frac{23}{7}, 3$$

$$y = 3, \frac{3}{2}$$

$$(10) x = 4, \frac{5}{3}$$

$$y = 2, \frac{7}{4}$$

$$(11) x = 2\sqrt{7}, \sqrt{7}$$

$$y = 7\sqrt{7}, 2\sqrt{7}$$

$$(12) x = 3, \frac{23}{11}$$

$$y = \frac{7}{2}, 3$$

$$(13) x = 5, 4$$

$$y = \frac{5}{2}, \frac{3}{2}$$

$$(14) x = 3, \frac{3}{2}$$

$$y = \frac{11}{3}, 2$$

$$(15) x = 4, 2$$

$$y = 2, \frac{11}{7}$$

$$(16) x = 2, \frac{8}{5}$$

$$y = 3, \frac{5}{2}$$

$$(17) x = 6\sqrt{2}, \sqrt{2}$$

$$y = 7\sqrt{2}, 2\sqrt{2}$$

$$(18) x = 2, \frac{9}{5}$$

$$y = \frac{33}{7}, 2$$

$$(19) x = \frac{9}{2}, 3$$

$$y = \frac{29}{11}, 2$$

$$(20) x = \frac{5}{2}, 2$$

$$y = 4, \frac{5}{3}$$

4. WRONG NUMBER SERIES

(1) 4096, 4079, 4045, 3990, 3926, 3841

- (a) 4079
- (b) 3990
- (c) 3926
- (d) 3841
- (e) None of these

(2) 134, 277, 846, 3397, 17052, 102345

- (a) 17052
- (b) 277
- (c) 846
- (d) 3397
- (e) None of these

(3) 65860, 32946, 11020, 2770, 576, 120

- (a) 11020
- (b) 32946
- (c) 576
- (d) 120
- (e) None of these

(4) 2450, 2474, 2509, 2563, 2620, 2700

- (a) 2474
- (b) 2700
- (c) 2563
- (d) 2620
- (e) None of these

(5) 156, 304, 368, 1082, 1188, 4736

- (a) 304
- (b) 1188
- (c) 368

(d) 4736

(e) None of these

(6) 140, 165, 294, 430, 698, 1091

- (a) 430
- (b) 294
- (c) 1091
- (d) 165

(e) None of these

(7) 72, 433, 2169, 8685, 26031, 52167

- (a) 8685
- (b) 2169
- (c) 26031
- (d) 433
- (e) None of these

(8) 850.75, 858.25, 869.25, 883.75, 901.25, 923.25

- (a) 858.25
- (b) 901.25
- (c) 923.25
- (d) 883.75
- (e) None of these

(9) 180, 294, 448, 720, 900, 1210

- (a) 448
- (b) 900
- (c) 294
- (d) 720
- (e) None of these

(10) 300, 323, 360, 420, 508, 674

- (a) 323
- (b) 674
- (c) 508

- (d) 360
(e) None of these
- (11) 46104, 7699, 1552, 405, 154, 100
(a) 46104
(b) 1552
(c) 154
(d) 100
(e) None of these
- (12) 400, 411, 438, 492, 620, 871
(a) 492
(b) 620
(c) 411
(d) 871
(e) None of these
- (13) 700, 715, 691, 726, 678, 731
(a) 726
(b) 731
(c) 691
(d) 715
(e) None of these
- (14) 85, 253, 751, 2225, 6673, 19969
(a) 2225
(b) 253
(c) 6673
(d) 751
(e) None of these
- (15) 150, 271, 440, 725, 1090, 1619
(a) 271
(b) 440
(c) 1619
(d) 725
(e) None of these
- (16) 125, 143, 185, 275, 455, 839
(a) 143
(b) 275
(c) 839

- (d) 185
(e) None of these
- (17) 75, 210, 155, 405, 315, 819
(a) 210
(b) 315
(c) 405
(d) 75
(e) None of these
- (18) 3200, 3172, 3129, 3060, 2986, 2880
(a) 2986
(b) 3060
(c) 3200
(d) 3129
(e) None of these
- (19) 190, 386, 1180, 2358, 7098, 14226
(a) 1180
(b) 386
(c) 2358
(d) 14226
(e) None of these
- (20) 820, 868, 940, 1048, 1200, 1453
(a) 1048
(b) 940
(c) 1453
(d) 1200
(e) None of these

Answers

- (1) b
(2) d
(3) a
(4) c

- (5) e
 (6) a
 (7) c
 (8) b
 (9) d
 (10) c
 (11) e
 (12) a
 (13) b
 (14) a
 (15) d
 (16) e
 (17) c
 (18) b
 (19) a
 (20) d

Solutions

- (1) $-17, -34, -51, -68, -85$.
 (2) $\times 2 + 9, \times 3 + 15, \times 4 + 21, \times 5 + 27, \times 6 + 33$.
 (3) $\div 2 + 16, \div 3 + 18, \div 4 + 20, \div 5 + 22, \div 6 + 24$.
 (4) $+24, +35, +48, +63, +80$; second differences: $+11, +13, +15, +17$.
 (5) $\times 2 - 8, +64, \times 3 - 12, +96, \times 4 - 16$.
 (6) Previous two terms $-11, -22, -33, -44$.
 (7) $\times 6 + 1^2, \times 5 + 2^2, \times 4 + 3^2, \times 3 + 4^2, \times 2 + 5^2$.
 (8) $+7.5, +11, +14.5, +18, +21.5$; second differences: $+3.5, +3.5, +3.5, +3.5$.
 (9) $6^3 - 6^2, 7^3 - 7^2, 8^3 - 8^2, 9^3 - 9^2, 10^3 - 10^2, 11^3 - 11^2$.
 (10) $+23, +37, +60, +97, +157$; each next difference = sum of previous two differences.
 (11) $\div 6 + 11, \div 5 + 13, \div 4 + 17, \div 3 + 19, \div 2 + 23$.
 (12) $+11, +27, +59, +123, +251$; each next difference = $\times 2 + 5$.
 (13) $+(3 \times 5), -(4 \times 6), +(5 \times 7), -(6 \times 8), +(7 \times 9)$.
 (14) $\times 3 - 2 \times 1^2, \times 3 - 2 \times 2^2, \times 3 - 2 \times 3^2, \times 3 - 2 \times 4^2, \times 3 - 2 \times 5^2$.
 (15) $+11^2, +13^2, +17^2, +19^2, +23^2$.
 (16) $+18, +42, +90, +186, +378$; each next difference = $\times 2 + 6$.
 (17) Odd places: $\times 2 + 5$; Even places: $\times 2 - 7$.
 (18) $-28, -43, -61, -82, -106$; second differences: $-15, -18, -21, -24$.
 (19) $\times 2 + 6, \times 3 + 12, \times 2 + 18, \times 3 + 24, \times 2 + 30$.
 (20) $+48, +72, +108, +162, +243$; each next difference = $\times 1.5$.

5. MISSING NUMBER SERIES

(1) 6200, 6177, 6131, ?, 5855, 5487

- (a) 6033
- (b) 6037
- (c) 6039
- (d) 6043
- (e) None of these

(2) 6, 17, 58, ?, 1216, 7309

- (a) 237
- (b) 241
- (c) 245
- (d) 249
- (e) None of these

(3) 28, 48, 84, ?, 220, 328

- (a) 132
- (b) 136
- (c) 138
- (d) 140
- (e) None of these

(4) 324, 342, 378, ?, 504, 594

- (a) 426
- (b) 428
- (c) 432
- (d) 436
- (e) None of these

(5) 7492, 3759, ?, 325, 72, 17

- (a) 1264
- (b) 1268

(c) 1272

- (d) 1276
- (e) None of these

(6) 86.5, 91.5, 99, ?, 121.5, 136.5

- (a) 107
- (b) 109
- (c) 111
- (d) 113
- (e) None of these

(7) 92, 173, 333, ?, 1279, 2535

- (a) 643
- (b) 645
- (c) 647
- (d) 649
- (e) None of these

(8) 5000, 4982, 4928, ?, 4280, 2822

- (a) 4758
- (b) 4762
- (c) 4766
- (d) 4770
- (e) None of these

(9) 35, 56, 96, ?, 249, 370

- (a) 159
- (b) 161
- (c) 163
- (d) 165
- (e) None of these

(10) 12, 33, 80, ?, 382, 793

- (a) 173
- (b) 175
- (c) 177
- (d) 179
- (e) None of these

(11) 148, 163, 193, ?, 373, 613

- (a) 249
- (b) 251
- (c) 253
- (d) 255
- (e) None of these

(12) 875, 868, 849, ?, 751, 660

- (a) 808
- (b) 812
- (c) 816
- (d) 820
- (e) None of these

(13) 24, 61, 122, ?, 340, 509

- (a) 209
- (b) 211
- (c) 213
- (d) 215
- (e) None of these

(14) 3546, 1782, ?, 180, 72, 57

- (a) 604
- (b) 608
- (c) 610
- (d) 612
- (e) None of these

(15) 225, 242, 276, ?, 480, 752

- (a) 342
- (b) 344
- (c) 346

- (d) 348
- (e) None of these

(16) 13, 31, 69, ?, 305, 623

- (a) 147
- (b) 149
- (c) 151
- (d) 153
- (e) None of these

(17) 218, 234, 259, ?, 344, 408

- (a) 291
- (b) 293
- (c) 295
- (d) 297
- (e) None of these

(18) 33, 64, 125, ?, 487, 968

- (a) 246
- (b) 248
- (c) 250
- (d) 252
- (e) None of these

(19) 1260, 1272, 1308, ?, 1740, 2712

- (a) 1408
- (b) 1410
- (c) 1412
- (d) 1416
- (e) None of these

(20) 57, 78, 116, ?, 251, 352

- (a) 171
- (b) 173
- (c) 175
- (d) 177
- (e) None of these

Answers:

- (1) c
- (2) b
- (3) d
- (4) c
- (5) a
- (6) b
- (7) d
- (8) c
- (9) a
- (10) d
- (11) c
- (12) b
- (13) c
- (14) d
- (15) b
- (16) a
- (17) c
- (18) a
- (19) d
- (20) b

Solutions:

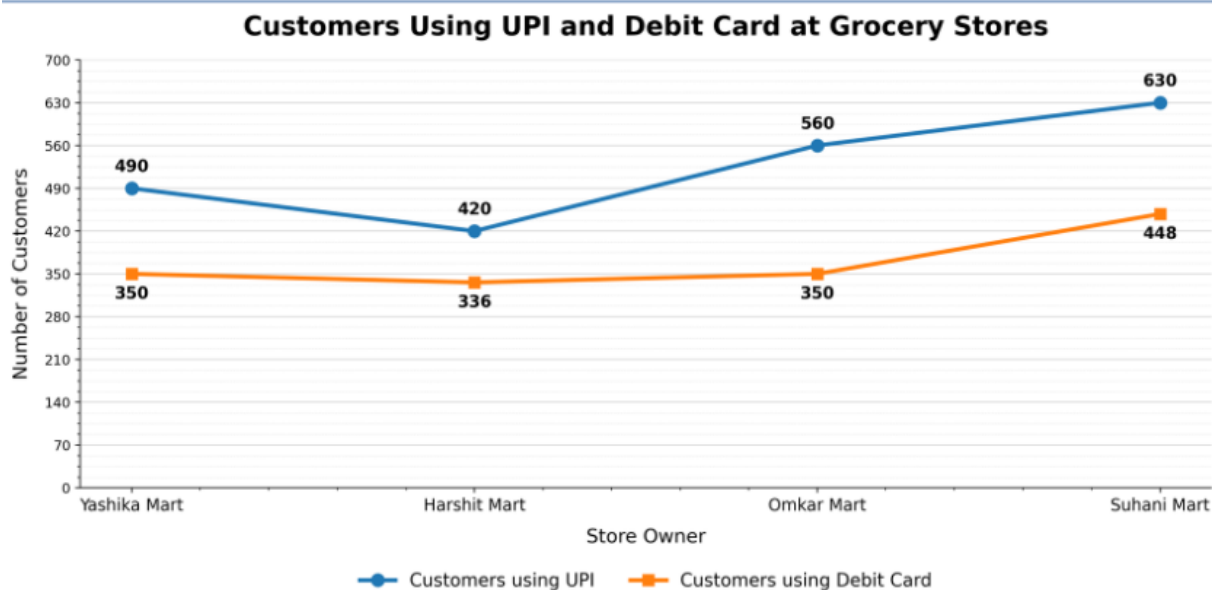
- (1) -23, -46, -92, -184, -368
- (2) $\times 2 + 5$, $\times 3 + 7$, $\times 4 + 9$, $\times 5 + 11$, $\times 6 + 13$
- (3) $+2 \times 10$, $+3 \times 12$, $+4 \times 14$, $+5 \times 16$, $+6 \times 18$
- (4) +18, +36, +54, +72, +90
- (5) $\div 2 + 13$, $\div 3 + 11$, $\div 4 + 9$, $\div 5 + 7$, $\div 6 + 5$

- (6) +5, +7.5, +10, +12.5, +15
- (7) $\times 2 - 11$, $\times 2 - 13$, $\times 2 - 17$, $\times 2 - 19$, $\times 2 - 23$
- (8) -18, -54, -162, -486, -1458
- (9) $+7 \times 3$, $+8 \times 5$, $+9 \times 7$, $+10 \times 9$, $+11 \times 11$
- (10) $\times 2 + 9$, $\times 2 + 14$, $\times 2 + 19$, $\times 2 + 24$, $\times 2 + 29$
- (11) +15, +30, +60, +120, +240
- (12) -7, -19, -37, -61, -91
Second difference: -12, -18, -24, -30
- (13) Terms are: $n^3 - 3$, for $n = 3, 4, 5, 6, 7, 8$
- (14) $\div 2 + 9$, $\div 3 + 18$, $\div 4 + 27$, $\div 5 + 36$, $\div 6 + 45$
- (15) +17, +34, +68, +136, +272
- (16) $\times 2 + 5$, $\times 2 + 7$, $\times 2 + 9$, $\times 2 + 11$, $\times 2 + 13$
- (17) $+4^2$, $+5^2$, $+6^2$, $+7^2$, $+8^2$
- (18) $\times 2 - 2$, $\times 2 - 3$, $\times 2 - 4$, $\times 2 - 5$, $\times 2 - 6$
- (19) +12, +36, +108, +324, +972
- (20) +21, +38, +57, +78, +101
Second difference: +17, +19, +21, +23

6. DATA INTERPRETATION

Set 1: The line graph shows the number of customers (male + female) who used UPI and the number of customers (male + female) who used debit cards for payment at four different grocery stores: Yashika Mart, Harshit Mart, Omkar Mart and Suhani Mart. Study the line graph carefully and answer the following questions.

रेखा आलेख चार अलग-अलग किराना स्टोरों: यशिका मार्ट, हर्षित मार्ट, ओमकार मार्ट और सुहानी मार्ट में UPI से भुगतान करने वाले ग्राहकों (पुरुष + महिला) और डेबिट कार्ड से भुगतान करने वाले ग्राहकों (पुरुष + महिला) की संख्या को दर्शाता है। रेखा आलेख का ध्यानपूर्वक अध्ययन करें और निम्नलिखित प्रश्नों के उत्तर दें।



Q1: Find the difference between the average number of customers who used UPI at Yashika Mart and Harshit Mart and the number of customers who used debit cards at Omkar Mart.

यशिका मार्ट और हर्षित मार्ट में UPI से भुगतान करने वाले ग्राहकों की औसत संख्या और ओमकार मार्ट में डेबिट कार्ड से भुगतान करने वाले ग्राहकों की संख्या के बीच अंतर ज्ञात कीजिए।

(A) 112

(B) 105

- (C) 98
- (D) 126
- (E) 84

Q2: In Harshit Mart, 75% of the customers who used debit cards were males. This number is 168 less than the number of male customers who used UPI at Omkar Mart. Find the number of female customers who used UPI at Omkar Mart.

हर्षित मार्ट में डेबिट कार्ड से भुगतान करने वाले ग्राहकों में से 75% पुरुष थे। यह संख्या ओमकार मार्ट में UPI से भुगतान करने वाले पुरुष ग्राहकों की संख्या से 168 कम है। ओमकार मार्ट में UPI से भुगतान करने वाली महिला ग्राहकों की संख्या ज्ञात कीजिए।

- (A) 160
- (B) 112
- (C) 126
- (D) 154
- (E) 140

Q3: The ratio of the total number of customers who used UPI at Yashika Mart and Suhani Mart together to the number of customers who used debit cards at Omkar Mart is $x : y$. Find the value of $x + y$.

यशिका मार्ट और सुहानी मार्ट में मिलकर UPI से भुगतान करने वाले ग्राहकों की कुल संख्या और ओमकार मार्ट में डेबिट कार्ड से भुगतान करने वाले ग्राहकों की संख्या का अनुपात $x : y$ है। $x + y$ का मान ज्ञात कीजिए।

- (A) 18
- (B) 24
- (C) 19
- (D) 21
- (E) 23

Q4: If every customer at each store used only one of the two given payment modes, then find the maximum total number of customers among the four stores.

यदि प्रत्येक स्टोर में हर ग्राहक ने दिए गए दो भुगतान माध्यमों में से केवल एक ही माध्यम का उपयोग किया, तो चारों स्टोरों में ग्राहकों की अधिकतम कुल संख्या ज्ञात कीजिए।

- (A) 1078
- (B) 910
- (C) 994
- (D) 840

(E) 756

Q5: Find the sum of the number of customers who used debit cards at Yashika Mart and Harshit Mart together and the number of customers who used UPI at Omkar Mart and Suhani Mart together.

यशिका मार्ट और हर्षित मार्ट में मिलकर डेबिट कार्ड से भुगतान करने वाले ग्राहकों की संख्या तथा ओमकार मार्ट और सुहानी मार्ट में मिलकर UPI से भुगतान करने वाले ग्राहकों की संख्या का योग ज्ञात कीजिए।

(A) 1764

(B) 1813

(C) 1876

(D) 1946

(E) 1708

Solutions:

Store Owner	Hindi Name	UPI Customers	Debit Card Customers	Total Customers
Yashika Mart	यशिका मार्ट	490	350	840
Harshit Mart	हर्षित मार्ट	420	336	756
Omkar Mart	ओमकार मार्ट	560	350	910
Suhani Mart	सुहानी मार्ट	630	448	1078

1. Option B

UPI customers at Yashika Mart = 490

UPI customers at Harshit Mart = 420

Average = $(490 + 420) \div 2 = 455$

Debit card customers at Omkar Mart = 350

Difference = $455 - 350 = 105$

2. Option E

Debit card customers at Harshit Mart = 336

Male debit card customers at Harshit Mart = $336 \times 75\% = 252$ Male UPI customers at Omkar Mart = $252 + 168 = 420$

Total UPI customers at Omkar Mart = 560

Female UPI customers at Omkar Mart = $560 - 420 = 140$

3. Option D

UPI customers at Yashika Mart and Suhani Mart = $490 + 630 = 1120$

Debit card customers at Omkar Mart = 350

Required ratio = $1120 : 350 = 16 : 5$

$x + y = 16 + 5 = 21$

4. Option A

Total customers at Yashika Mart = $490 + 350 = 840$

Total customers at Harshit Mart = $420 + 336 = 756$

Total customers at Omkar Mart = $560 + 350 = 910$

Total customers at Suhani Mart = $630 + 448 = 1078$

Maximum total customers = 1078

5. Option C

Debit card customers at Yashika Mart and Harshit Mart = $350 + 336 = 686$

UPI customers at Omkar Mart and Suhani Mart = $560 + 630 = 1190$

Required sum = $686 + 1190 = 1876$

Set 2: The table shows the total number of visitors (male + female) who attended a Science Fair on five different days of a week. It also shows the ratio of male to female visitors on the respective days. Read the table carefully and answer the questions.

तालिका सप्ताह के पाँच अलग-अलग दिनों में विज्ञान मेले में आए कुल आगंतुकों (पुरुष + महिला) की संख्या दर्शाती है। यह संबंधित दिनों में पुरुष और महिला आगंतुकों का अनुपात भी दर्शाती है। तालिका को ध्यानपूर्वक पढ़िए और प्रश्नों के उत्तर दीजिए।

Day / दिन	Total visitors at Science Fair / विज्ञान मेले में कुल आगंतुक	Ratio of male to female visitors / पुरुष और महिला आगंतुकों का अनुपात
Monday / सोमवार	1584	5:3
Tuesday / मंगलवार	1848	8:3
Wednesday / बुधवार	1320	2:3
Thursday / गुरुवार	1760	3:1
Friday / शुक्रवार	1980	5:1

Q1: If the ratio of female visitors on Tuesday to female visitors on Wednesday is $P:Q$, then find $P + Q$.

यदि मंगलवार को महिला आगंतुकों और बुधवार को महिला आगंतुकों का अनुपात $P:Q$ है, तो $P + Q$ ज्ञात कीजिए।

- (A) 16
- (B) 22
- (C) 20
- (D) 18
- (E) 24

Q2: The number of male visitors on Friday is what percentage more or less than the number of male visitors on Thursday?

शुक्रवार को पुरुष आगंतुकों की संख्या, गुरुवार को पुरुष आगंतुकों की संख्या से कितने प्रतिशत अधिक या कम है?

- (A) 20%
- (B) 25%
- (C) 33.33%
- (D) 16.67%
- (E) 40%

Q3: The number of male visitors on Saturday is 20% more than the number of male visitors on Monday. If the ratio of male to female visitors on Saturday is 9:10, then find the number of female visitors on Saturday.

शनिवार को पुरुष आगंतुकों की संख्या, सोमवार को पुरुष आगंतुकों की संख्या से 20% अधिक है। यदि शनिवार को पुरुष और महिला आगंतुकों का अनुपात 9:10 है, तो शनिवार को महिला आगंतुकों की संख्या ज्ञात कीजिए।

- (A) 1260
- (B) 1288
- (C) 1350
- (D) 1296
- (E) 1320

Q4: If the total visitors on Monday, Tuesday and Wednesday are shown in a pie chart, then find the central angle made by the visitors on Monday.

यदि सोमवार, मंगलवार और बुधवार के कुल आगंतुकों को एक पाई चार्ट में दर्शाया जाए, तो सोमवार के आगंतुकों द्वारा बनाया गया केंद्रीय कोण ज्ञात कीजिए।

- (A) 120°
- (B) 108°
- (C) 135°
- (D) 144°
- (E) 126°

Q5: 75% of the female visitors who attended the Science Fair on Monday, Thursday and Friday together bought entry coupons. Find the total number of female visitors who bought entry coupons on these three days together.

सोमवार, गुरुवार और शुक्रवार को मिलाकर विज्ञान मेले में आई महिला आगंतुकों में से 75% ने प्रवेश कूपन खरीदे। इन तीनों दिनों में मिलाकर प्रवेश कूपन खरीदने वाली महिला आगंतुकों की कुल संख्या ज्ञात कीजिए।

- (A) 996
- (B) 1056
- (C) 1023
- (D) 1012
- (E) 1089

Solutions:

Day / दिन	Total visitors / कुल आगंतुक	Male visitors / पुरुष	Female visitors / महिला
Monday / सोमवार	1584	990	594
Tuesday / मंगलवार	1848	1344	504
Wednesday / बुधवार	1320	528	792
Thursday / गुरुवार	1760	1320	440
Friday / शुक्रवार	1980	1650	330

Common values:

Monday: $5 + 3 = 8$ parts, $1584/8 = 198$, Male = 990, Female = 594

Tuesday: $8 + 3 = 11$ parts, $1848/11 = 168$, Male = 1344, Female = 504

Wednesday: $2 + 3 = 5$ parts, $1320/5 = 264$, Male = 528, Female = 792

Thursday: $3 + 1 = 4$ parts, $1760/4 = 440$, Male = 1320, Female = 440

Friday: $5 + 1 = 6$ parts, $1980/6 = 330$, Male = 1650, Female = 330

1. Option D

Female visitors on Tuesday and Wednesday = 504:792

Ratio = 7:11

$P + Q = 7 + 11 = 18$

2. Option B

Male visitors on Friday = 1650

Male visitors on Thursday = 1320

Difference = $1650 - 1320 = 330$

Required percentage = $330/1320 \times 100 = 25\%$ more

3. Option E

Male visitors on Saturday = 120% of 990 = 1188

Male:Female on Saturday = 9:10

Female visitors on Saturday = $1188 \times 10/9 = 1320$

4. Option A

Total visitors on Monday, Tuesday and Wednesday = $1584 + 1848 + 1320 = 4752$

Central angle for Monday = $1584/4752 \times 360$

Central angle = 120°

5. Option C

Female visitors on Monday, Thursday and Friday = $594 + 440 + 330 = 1364$

Required value = 75% of 1364

Required value = $\frac{3}{4} \times 1364 = 1023$

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Set 3: The line chart shows the total number of tour packages sold, in hundreds, by five tour coordinators named Sarthak, Pranay, Tarun, Nandita and Vaidehi. It also shows the number of domestic tour packages sold by each of them as a percentage of their total tour packages sold.

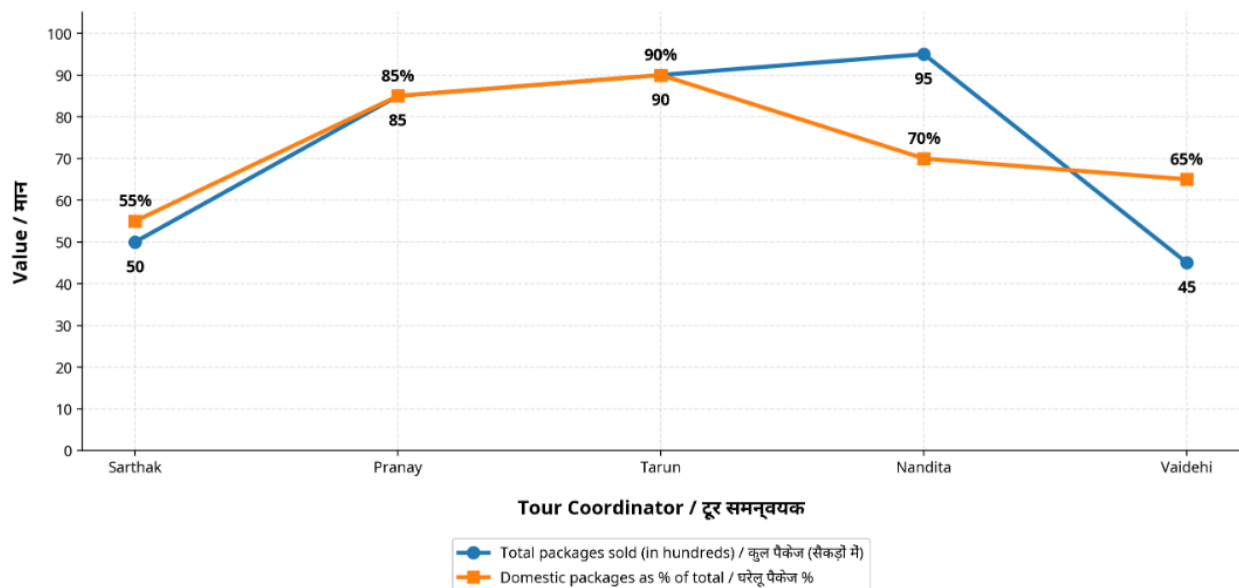
रेखा आलेख पांच टूर समन्वयकों सार्थक, प्रणय, तरुण, नंदिता और वैदेही द्वारा बेचे गए कुल टूर पैकेजों की संख्या, सैकड़ों में, दर्शाता है। यह प्रत्येक समन्वयक द्वारा बेचे गए घरेलू टूर पैकेजों की संख्या को उनके कुल बेचे गए टूर पैकेजों के प्रतिशत के रूप में भी दर्शाता है।

Note: Total packages sold = Domestic packages sold + International packages sold.

बेचे गए कुल पैकेज = बेचे गए घरेलू पैकेज + बेचे गए अंतरराष्ट्रीय पैकेज।

Tour Packages Sold: Total and Domestic Share

टूर पैकेज बिक्री: कुल और घरेलू हिस्सा



Q1: Find the average number of international tour packages sold by Sarthak, Nandita and Vaidehi.

सार्थक, नंदिता और वैदेही द्वारा बेचे गए अंतरराष्ट्रीय टूर पैकेजों की औसत संख्या ज्ञात कीजिए।

- (A) 2125
- (B) 2175
- (C) 2275
- (D) 2225
- (E) 2325

Q2: The number of international tour packages sold by Kunal is $83\frac{1}{3}\%$ of his total packages sold. This number is also $66\frac{2}{3}\%$ more than the number of domestic tour

packages sold by Tarun. Find the total number of tour packages sold by Kunal.

कुणाल द्वारा बेचे गए अंतरराष्ट्रीय टूर पैकेजों की संख्या उसके कुल बेचे गए पैकेजों की $83\frac{1}{3}\%$ है। यह संख्या तरुण द्वारा बेचे गए घरेलू टूर पैकेजों की संख्या से $66\frac{2}{3}\%$ अधिक भी है। कुणाल द्वारा बेचे गए कुल टूर पैकेजों की संख्या ज्ञात कीजिए।

- (A) 16200
- (B) 15800
- (C) 16800
- (D) 15400
- (E) 17200

Q3: The number of domestic tour packages sold by Sarthak and Pranay together is what percent more than the number of domestic tour packages sold by Nandita?

सार्थक और प्रणय द्वारा मिलाकर बेचे गए घरेलू टूर पैकेजों की संख्या, नंदिता द्वारा बेचे गए घरेलू टूर पैकेजों की संख्या से कितने प्रतिशत अधिक है?

- (A) 45%
- (B) 55%
- (C) 60%
- (D) 40%
- (E) 50%

Q4: The ratio of domestic tour packages sold by Vaidehi to international tour packages sold by Pranay and Tarun together is $x:y$. Find $x + y$.

वैदेही द्वारा बेचे गए घरेलू टूर पैकेजों और प्रणय तथा तरुण द्वारा मिलाकर बेचे गए अंतरराष्ट्रीय टूर पैकेजों का अनुपात $x:y$ है। $x + y$ ज्ञात कीजिए।

- (A) 64
- (B) 68
- (C) 72
- (D) 76
- (E) 84

Q5: Find the number of international tour packages sold by the coordinator who sold the highest number of domestic tour packages.

जिस समन्वयक ने सबसे अधिक घरेलू टूर पैकेज बेचे, उसके द्वारा बेचे गए अंतरराष्ट्रीय टूर पैकेजों की संख्या ज्ञात कीजिए।

- (A) 800
- (B) 750
- (C) 900

(D) 1050

(E) 1200

Solutions:

1. Option D

Sarthak total packages = $50 \times 100 = 5000$

Sarthak international packages = 45% of 5000 = 2250

Nandita total packages = $95 \times 100 = 9500$

Nandita international packages = 30% of 9500 = 2850

Vaidehi total packages = $45 \times 100 = 4500$

Vaidehi international packages = 35% of 4500 = 1575

Required average = $(2250 + 2850 + 1575)/3 = 2225$

2. Option A

Tarun total packages = $90 \times 100 = 9000$

Tarun domestic packages = 90% of 9000 = 8100

Kunal international packages = $166\frac{2}{3}\%$ of 8100 = $8100 \times \frac{5}{3} = 13500$

Kunal international packages = $83\frac{1}{3}\%$ of Kunal total = $\frac{5}{6}$ of Kunal total

Kunal total packages = $13500 \times \frac{6}{5} = 16200$

3. Option E

Sarthak domestic packages = 55% of 5000 = 2750

Pranay total packages = $85 \times 100 = 8500$

Pranay domestic packages = 85% of 8500 = 7225

Nandita domestic packages = 70% of 9500 = 6650

Required percentage = $(2750 + 7225 - 6650)/6650 \times 100$

Required percentage = $3325/6650 \times 100 = 50\%$

4. Option B

Vaidehi domestic packages = 65% of 4500 = 2925

Pranay international packages = 15% of 8500 = 1275

Tarun international packages = 10% of 9000 = 900

Required ratio = $2925:(1275 + 900)$

Required ratio = $2925:2175 = 39:29$

$x + y = 39 + 29 = 68$

5: Option C

Sarthak domestic packages = 2750

Pranay domestic packages = 7225

Tarun domestic packages = 8100

Nandita domestic packages = 6650

Vaidehi domestic packages = 2925

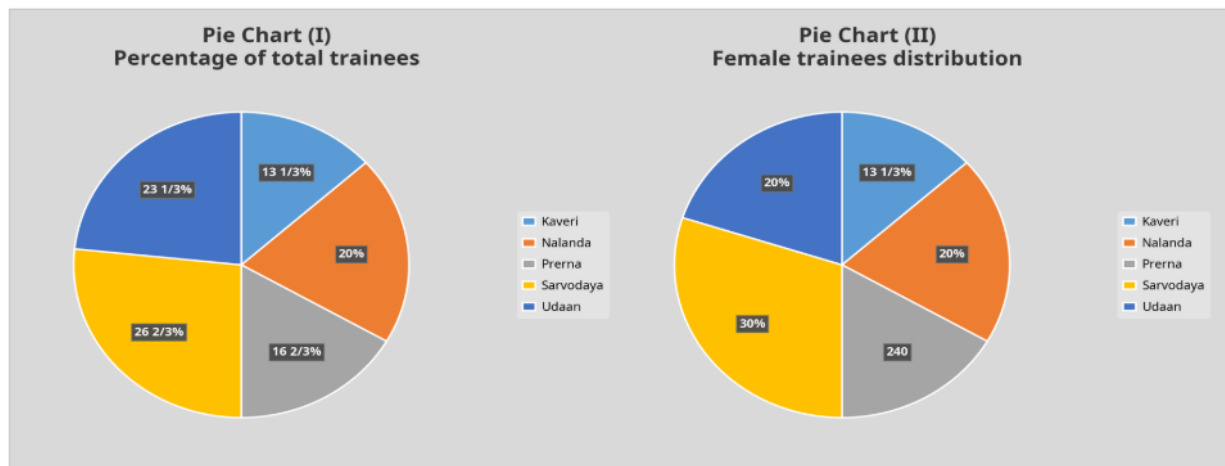
Highest domestic packages were sold by Tarun

Tarun international packages = 10% of 9000 = 900

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Set 4: Given below are two pie charts. Pie chart (I) shows the percentage distribution of total trainees, male and female together, in five cultural academies: Kaveri, Nalanda, Prerna, Sarvodaya and Udaan. Pie chart (II) shows the percentage distribution or number of female trainees in these academies. Total trainees in an academy = male trainees + female trainees. Male trainees in Prerna academy are 150% more than female trainees in Prerna academy. Read the data carefully and answer the following questions.

नीचे दो पाई चार्ट दिए गए हैं। पाई चार्ट (I) पाँच सांस्कृतिक अकादमियों, कावेरी, नालंदा, प्रेरणा, सर्वोदय और उड़ान, में कुल प्रशिक्षुओं, यानी पुरुष और महिला प्रशिक्षुओं को मिलाकर, का प्रतिशत वितरण दर्शाता है। पाई चार्ट (II) इन अकादमियों में महिला प्रशिक्षुओं का प्रतिशत वितरण या संख्या दर्शाता है। किसी अकादमी में कुल प्रशिक्षु = पुरुष प्रशिक्षु + महिला प्रशिक्षु। प्रेरणा अकादमी में पुरुष प्रशिक्षु, प्रेरणा अकादमी की महिला प्रशिक्षुओं से 150% अधिक हैं। आंकड़ों को ध्यान से पढ़िए और निम्नलिखित प्रश्नों के उत्तर दीजिए।



Q1: The total number of trainees in Nalanda academy is what percentage of the number of male trainees in Prerna academy?

नालंदा अकादमी में कुल प्रशिक्षुओं की संख्या, प्रेरणा अकादमी में पुरुष प्रशिक्षुओं की संख्या का कितना प्रतिशत है?

- (A) 150%
- (B) 168%
- (C) 175%
- (D) 160%
- (E) 180%

Q2: Total female trainees in Kaveri, Nalanda and Prerna academies together are what percentage more than the male trainees in Prerna academy?

कावेरी, नालंदा और प्रेरणा अकादमियों में मिलाकर कुल महिला प्रशिक्षु, प्रेरणा अकादमी के

पुरुष प्रशिक्षुओं से कितने प्रतिशत अधिक हैं?

- (A) 15%
- (B) 25%
- (C) 30%
- (D) 20%
- (E) 35%

Q3: Find the difference between male trainees in Kaveri and Sarvodaya academies together and female trainees in Prerna academy.

कावेरी और सर्वोदय अकादमियों में मिलाकर पुरुष प्रशिक्षुओं और प्रेरणा अकादमी में महिला प्रशिक्षुओं के बीच का अंतर ज्ञात कीजिए।

- (A) 1152
- (B) 1080
- (C) 1200
- (D) 1128
- (E) 1188

Q4: Find the ratio of total trainees in Kaveri and Sarvodaya academies together to female trainees in Nalanda and Udaan academies together.

कावेरी और सर्वोदय अकादमियों में मिलाकर कुल प्रशिक्षुओं का नालंदा और उड़ान अकादमियों में मिलाकर महिला प्रशिक्षुओं से अनुपात ज्ञात कीजिए।

- (A) 5:2
- (B) 9:4
- (C) 11:5
- (D) 13:4
- (E) 7:2

Q5: Find the difference between the total number of male trainees and the total number of female trainees in all five academies together.

पाँचों अकादमियों में कुल पुरुष प्रशिक्षुओं और कुल महिला प्रशिक्षुओं की संख्या के बीच का अंतर ज्ञात कीजिए।

- (A) 2040
- (B) 2280
- (C) 2160
- (D) 1980
- (E) 2400

Solutions:

Academy	Total trainees	Female trainees	Male trainees	Pie (I) label	Pie (II) label
Kaveri	672	192	480	13 1/3%	13 1/3%
Nalanda	1008	288	720	20%	20%
Prerna	840	240	600	16 2/3%	240
Sarvodaya	1344	432	912	26 2/3%	30%
Udaan	1176	288	888	23 1/3%	20%
Total	5040	1440	3600	100%	100%

Common Calculation:

Female trainees in Prerna = 240

Prerna female percentage = $16 \frac{2}{3}\% = \frac{1}{6}$

Total female trainees = $240 \times 6 = 1440$

Male trainees in Prerna = 250% of 240 = 600

Total trainees in Prerna = $240 + 600 = 840$

Prerna total percentage = $16 \frac{2}{3}\% = \frac{1}{6}$

Total trainees in all academies = $840 \times 6 = 5040$

Total trainees: Kaveri = 672, Nalanda = 1008, Prerna = 840, Sarvodaya = 1344, Udaan = 1176

Female trainees: Kaveri = 192, Nalanda = 288, Prerna = 240, Sarvodaya = 432, Udaan = 288

Male trainees: Kaveri = 480, Nalanda = 720, Prerna = 600, Sarvodaya = 912, Udaan = 888

1. Option B

Total trainees in Nalanda = 1008

Male trainees in Prerna = 600

Required percentage = $1008/600 \times 100 = 168\%$

2. Option D

Female trainees in Kaveri, Nalanda and Prerna = $192 + 288 + 240 = 720$

Male trainees in Prerna = 600

Difference = $720 - 600 = 120$

Required percentage = $120/600 \times 100 = 20\%$

3. Option A

Male trainees in Kaveri = 480

Male trainees in Sarvodaya = 912

Female trainees in Prerna = 240

Required difference = $480 + 912 - 240 = 1152$

4. Option E

Total trainees in Kaveri and Sarvodaya = $672 + 1344 = 2016$

Female trainees in Nalanda and Udaan = $288 + 288 = 576$

Required ratio = $2016:576 = 7:2$

5. Option C

Total male trainees = $480 + 720 + 600 + 912 + 888 = 3600$

Total female trainees = 1440

Required difference = $3600 - 1440 = 2160$